



EAST AFRICA UNIVERSITY – GAROWE

Faculty of Technology and Engineering

THESIS PROJECT GUIDELINE

This document provides comprehensive guidelines for final-year students in the Department of Computer Science and Telecommunication Engineering.

1. OBJECTIVES OF THE THESIS

- Apply theoretical knowledge to real-world problems
- Develop practical systems or prototypes
- Conduct academic and scientific research
- Enhance problem-solving, analytical, and innovation skills

2. TYPES OF THESIS

A. Computer Science

- Web and Mobile Applications
- Artificial Intelligence / Machine Learning Systems
- Database Systems
- Cybersecurity Applications
- Cloud-Based Systems

B. Telecommunication Engineering

- Network Design and Simulation
- Wireless Communication Systems
- Internet of Things (IoT) Systems
- Embedded Communication Devices
- Signal Processing Systems

3. THESIS COMPONENTS

A. Practical Project

- System Design (UML diagrams, architecture diagrams)
- Implementation (software coding or hardware development)
- Testing and Validation

B. Research Work

- Literature Review
- Problem Analysis
- Methodology
- Results and Discussion

4. THESIS STRUCTURE

Chapter 1: Introduction

- Background of the Study
- Problem Statement
- Objectives (General & Specific)
- Scope of the Study
- Significance of the Study

Chapter 2: Literature Review

- Review of Related Studies
- Analysis of Existing Systems
- Identification of Research Gap

Chapter 3: Methodology

- Research Design
- Tools and Technologies Used
- System Architecture / Framework

Chapter 4: System Design and Implementation

- System Diagrams (Use Case, ERD, Network Diagrams)
- Development Process
- User Interface or Hardware Design

Chapter 5: Results and Discussion

- Testing Results
- System Performance Analysis
- Interpretation of Findings

Chapter 6: Conclusion and Recommendations

- Summary of Findings
- Conclusions
- Recommendations for Future Work

5. TECHNICAL REQUIREMENTS

Computer Science

- Use of programming languages (Python, Java, JavaScript, C#, etc.)
- Database integration (MySQL, PostgreSQL, MongoDB, etc.)
- Version control (GitHub or GitLab repository required)
- Deployment of system (web hosting or cloud platform preferred)

Telecommunication Engineering

- Use of simulation tools (Cisco Packet Tracer, MATLAB, NS2/NS3)
- Hardware implementation (where applicable)
- Network/system performance evaluation

6. EVALUATION CRITERIA

Component	Weight
Problem Definition	10%
Literature Review	15%
Methodology	15%
Implementation	25%
Results & Analysis	15%
Documentation	10%
Presentation	10%

7. DELIVERABLES

- Printed thesis document (hardcopy)
- Softcopy (PDF + source code)
- Functional system or prototype
- Presentation slides (PowerPoint or equivalent)

8. TIMELINE

Activity	Timeline
Proposal Submission	
Mid Evaluation	
Final Submission	
Final Defense	

9. ETHICAL CONSIDERATIONS

- Strictly avoid plagiarism
- Properly cite all sources
- Ensure originality of work
- Maintain academic integrity and honesty

10. REFERENCING STYLE (APA 7th EDITION)

All students must use **APA 7th edition** for citations and references.

In-text Citation Examples:

- One author: (Smith, 2020)
- Two authors: (Smith & Ali, 2021)
- Three or more: (Smith et al., 2022)

Reference List Examples:

Book:

Smith, J. (2020). *Introduction to computer systems*. Oxford University Press.

Journal Article:

Ali, M., & Hassan, A. (2021). Network optimization techniques. *International Journal of Technology*, 12(3), 45–60.

Website:

World Bank. (2022). Digital development overview. <https://www.worldbank.org>

11. APPENDICES

Appendices contain supplementary materials that support the thesis but are not included in the main chapters.

Examples:

- Questionnaire or Survey Instruments
- Interview Questions
- Source Code Snippets
- System Screenshots
- Network Configurations
- User Manuals

Each appendix should be labeled clearly:

- Appendix A
- Appendix B
- Appendix C